

CO 2 - AT 3

Class test

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Concept of image Enhancement in
Domain.

~~Spatial~~

Techniques of Spatial Domain enhancement

- Gray - level Transformation
- * Changes the brightness and Contrast of image
- * Examples: Negative transformation, log transformation, Gamma transformation
- * Application: Medical image, X-ray

~~Histogram Processing~~

- A histogram shows the distribution of pixel intensity values.
- Histogram Equalization improves contrast by spreading the intensity values.
- Application: Satellite image, fingerprint

3. Spatial Filtering

- Uses a small matrix (kernel) to modify pixel values.
- Smoothing filters remove noise
- Sharpening filters enhance edges and

Differentiate between image smoothing and image sharpening.

Image smoothing and image sharpening are important image enhancement techniques. Smoothing removes noise, while sharpening improves image detail.

Image Smoothing

Reduces noise by averaging neighbouring pixels.

Produces a smoother image.

Examples: Mean filter, Gaussian filter, Laplacian filter.

Application:

Medical imaging

Photography

Q. Explain Image Enhancement in the Frequency domain

Frequency domain enhancement modifies the frequency components of an image instead of directly changing pixels. It uses the Fourier transform.

Low-pass filter (LPF).

- Removes high-frequency noise.
- Produce smooth images.
- Used for noise reduction.

High-pass filter (HPF)

- Enhance high-frequency components edges.
- Produces sharp images.
- Used for image detection.